

SmartOps AI – Test Cases Document

Module 1 – Chaos Demo Platform

TC-01 DB Timeout Injection
TC-02 Memory Leak Injection
TC-03 CPU Spike Injection
TC-04 Missing ENV Injection
TC-05 Logs API Validation

Module 2 – GitHub Agent

TC-06 Valid Repository Analysis
TC-07 Invalid Repository Handling
TC-08 Commit Analysis Validation

Module 3 – Log Agent

TC-09 Runtime Log Retrieval
TC-10 Platform Failure Handling

Module 4 – Investigation Engine

TC-11 Timeline Generation
TC-12 Severity Classification

Module 5 – RAG Testing

TC-13 Similar Incident Retrieval
TC-14 Empty Knowledge Base Handling

Module 6 – Gemini RCA Testing

TC-15 RCA Generation
TC-16 Invalid Response Handling

Module 7 – Risk Engine

TC-17 Critical Incident Scoring
TC-18 Low Severity Scoring

Module 8 – PDF Generation

TC-19 PDF Download Validation
TC-20 Content Verification

Module 9 – Frontend Testing

TC-21 Form Validation
TC-22 Dashboard Loading
TC-23 Timeline Visualization
TC-24 Knowledge Graph Validation

End-to-End Testing

TC-25 Complete DB Timeout Flow
TC-26 Memory Leak Flow
TC-27 Missing ENV Flow

Review Demonstration Cases

TC-01, TC-06, TC-09, TC-15, and TC-25 are sufficient to demonstrate complete system functionality.